



Water film thickness measurement system for oil-water pipe flow

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ABSTRACT

A new technique development for high spatial and temporal resolution film-thickness measurement in oil-water flow is presented in this work. A capacitance measurement system is proposed to measure thin water films or local water fractions near to the pipe wall. The measurement system is based in a planar sensor which is flexible and can be placed close to and following the curvature of the inner wall of the tube. Experimental results of the capacitive technique showed that the system is able to measure film thickness between 400 μm and 2200 μm . Dispersed and core-annular flows were studied in a vertical 12-m-height glass pipe with 50.8 mm of internal diameter, using mineral oil (860 kg/m³ of density and 220 mPa s of viscosity) and tap water. Local time-domain signals and axial and circumferential distributions of water film thickness near to the pipe wall and 3-D images of water film are presented. The measured film thickness, obtained by the proposed technique, was compared with that obtained through a camera-based imaging technique and the agreement was good, with a deviation less than 20%. The new technique allows the observation and characterization of average interfacial topologies in separated flows, and high frequency phenomena related to dispersed flows that occur in liquid-liquid flow near the pipe wall.

1. Introduction

Consumption of energy is expected to continue increasing in future years. In coming decades, fossil fuels will still be the primary source of energy, with 78% of total world energy consumption [1]. Some of these energy resources are harder to be extracted at the present than in the past [2]. The production challenges are related, in part, to oil reservoirs location, i.e., most of the new fields are offshore, at great depths under the sea and in some cases, to the high oil viscosity. The heavy-oil recovery matters are: the high flow resistance, the pressure drop, the need of heating and dilution during transport, lower market value and cost increase. This scenario has created an interesting research opportunity, aiming to develop new technologies improving the heavy oil recovery factor [3]. Several efforts had been done to explain the utility of reducing the friction in pipes. For example, Rodriguez et al. [4] increased the production rate in a full scale on-shore experiment using core-annular flow. Rodriguez et al. [5] studied the drag reduction phenomenon in dispersed flows and proposed a model to predict it.

There is growing attention on methods and technologies to improve the heavy oil transportation process [6]. Particularly, in methods

focused on decreasing flow resistance to values low enough such that the pumping requirements and pipeline size reach viable values [7]. Lubricated transportation using core-annular flow and reverse emulsion (oil-in-water emulsion) have been recommended as techniques to reduce the pressure drop due to viscous effects [3,8]. Recently, reverse emulsion has been suggested as a promising solution to reduce the total pressure gradient and improve the gas-lift technique [9].

1.1. Core-annular flow

In core-annular flow, water flow is injected in the pipe in a way that an annular flow pattern is produced, with water near the wall surrounding an oil core. The water annulus keeps the oil from sticking to the pipe wall [3,10]. Therefore, the frictional pressure gradient is comparable to that observed in single-phase water flow, resulting in a significant decrease of pressure drop, but the more viscous the oil, the more intense the pressure-drop reduction [11]. According to experimental studies, for the core-annular flow pattern to occur the core phase should be much more viscous than the annulus phase and the in-situ volumetric fraction of the latter should be lower than 50%.

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